Supplement, appendix and contribution statement

Everything here supports the main document; none of it changes a registered result. Post-hoc and synthetic analyses are labelled as such. Model names: B1c = per-state quantile rule with the calibration shift; C* = linear quantile regression; D* = shallow HGB; B0 = the project's retained rule (no shift).

S1. Model selection on validation data (F0 → V)

Metric: pinball loss at q = 0.90 on the unshifted integer promise $\max\{1, \lceil \hat{y} \rceil\}$. A linear step was adopted only if it lowered the loss by at least 1%. Candidates within 1.001 × the global minimum counted as tied; inside a tie the simpler setting won.

Candidate	V loss	Change vs current best	Decision
C0 registered linear (log1p target, numeric weekday, no penalty)	1.28856	—	start
C1 weekday as a category	1.29170	−0.24%	rejected
C2 raw-days target	1.28932	−0.06%	rejected
C3 L1 penalty, best of 3 alphas (0.0001)	1.28713	+0.11%	rejected (< 1%)
D3 HGB, depth 2, 400 iterations (best of 8)	1.26266	—	selected; tie set D3, D4, D7

Post-hoc intervals on V. A week-block bootstrap on V (14 weeks, 2,000 draws) gives an improvement over C0 of 0.11% (−0.002% to 0.245%) for the C3 pick and 2.01% (1.11% to 2.85%) for D3. No improvement in the first interval reaches the pre-set 1% bar, so the original choice stands. Both candidates were picked on this same V, and the intervals leave out selection and refitting uncertainty, so D3's figure is optimistic and was only a reason to evaluate it.

S2. A small simulation of two calibration failures (synthetic data)

Question: does a calibration shift fail because the delivery distribution changes over time, or because at the freeze only the fast, already-delivered orders can be used? Delivery days follow $\log Y = 2.2 + 0.5x + 0.45\varepsilon$, rounded up. The predictor is the conditional median,

and the calibration code is the project's. Each of 1,000 replications draws 2,000 calibration and 5,000 evaluation orders (seed 20261021). All scenarios share the evaluation sample. For the time-change scenarios, the continuous delivery time is rescaled before rounding.

Scenario minus the stable, complete-label scenario	Coverage change, pts (mean ± MC s.e.)	Middle 90% of replications	Promised days change
Only delivered-by-freeze labels (90-day window)	−1.68 ± 0.015	−2.48 to −0.94	−0.99
Random subset of the same size (diagnostic)	+0.02 ± 0.009	−0.42 to +0.46	+0.02
Later deliveries 15% slower	−5.77 ± 0.014	−6.50 to −5.08	0
Later deliveries 15% faster	+4.26 ± 0.012	+3.66 to +4.90	0

The two mechanisms leave different traces. Selective labels shrink the shift itself, so promised days and coverage fall together; a random subset of the same size does not. A later change in delivery times leaves the promise unchanged and moves only coverage. The "middle 90%" column is the spread of replications, not a confidence interval for the mean. A first, unpaired run (seed 20261020) gave the same picture. This explains mechanisms; it does not show which one, if either, acted in the Olist data.

S3. Post-hoc sensitivity checks (added after the third review)

All tables below reuse the saved per-order predictions from the frozen models. They choose no policy and change no registered verdict. Source: scripts/e1_posthoc_review3.py → outputs/e1/posthoc_r3/ (posthoc_r3.json, SHA-256 5f7c2265…; checksums in MANIFEST.sha256).

S3a. Days needed at equal coverage on E. Each model's shift is swept over E in steps of 0.01 day; the table reads off the mean promised days at the first step reaching each coverage level. The state rule moves in whole-day steps, so it can only reach some levels by overshooting them.

Coverage on E	Linear C*	HGB D*	State rule	HGB − linear
90.00% (target)	17.20	16.60	17.99	−0.60
93.95% (B0′s coverage)	19.83	19.12	20.99	−0.71
96.18% (C* registered)	22.44	21.71	23.99	−0.73
97.10% (D* registered)	24.57	23.66	25.99	−0.91

On C at the registered shifts: D* 91.14% coverage with 24.54 days, C* 91.19% with 24.68, B1c 90.30% with 24.49.

S3b. Integer-aware calibration shift. The registered shift is the k-th smallest residual $y - \hat{y}$, followed by rounding up. For a continuous $\hat{y}$ this leaves up to one day of slack. Taking instead the k-th smallest of $y - 1 - \hat{y}$ lowers the shift by exactly 1 for a continuous predictor and leaves an integer predictor (the state rule) unchanged.

Policy	Shift: registered → integer-aware	Coverage on C	Days on E	Coverage on E
B1c state rule	6.00 → 6.00	90.30% → 90.30%	23.99 → 23.99	96.49% → 96.49%
B2 earlier regression	6.02 → 5.02	91.17% → 90.00%	22.87 → 21.87	96.45% → 95.87%
C* linear	6.15 → 5.15	91.19% → 90.00%	22.45 → 21.45	96.18% → 95.51%
D* HGB	6.46 → 5.46	91.14% → 90.00%	23.67 → 22.67	97.10% → 96.66%

With the integer-aware shift, D* would be 1.32 days shorter than B1c at +0.18 points of coverage, and C* 2.54 days shorter at −0.97 points. No interval was computed, and this formula was not registered; the registered H3 verdict stands. A fair test belongs in a new, separately registered study.

S3c. The evaluation window. The project′s population keeps orders whose Olist estimated delivery week is no later than 2018-07-23. Late in the window, an order enters only if Olist promised it a short time.

Purchase month (E)	Orders	Mean actual days	Olist promise, mean	Olist promise, max	Late vs Olist
April 2018	6,670	11.47	24.27	63	4.59%
May 2018	6,630	11.36	23.41	60	6.67%

Purchase month (E)	Orders	Mean actual days	Olist promise, mean	Olist promise, max	Late vs Olist
June 2018	5,565	8.92	27.45	56	1.26%
July 2018	1,085	6.22	16.33	26	2.95%

Policy	April–May coverage	April–May days	June–July coverage	June–July days
B0 retained rule	92.23%	20.55	97.38%	19.82
B1c state rule	95.53%	24.15	98.39%	23.67
C* linear	95.13%	22.75	98.27%	21.85
D* HGB	96.39%	24.01	98.51%	22.99

The truncated months explain part of the over-coverage, but April–May alone is still well above 90% for every calibrated policy.

S3d. Calibration season. Late rates against Olist's date on C, by purchase month: October 2017 4.24% (4,414 orders), November 12.49% (7,191; 42% of C), December 7.43% (5,451). The C shift was therefore learned mostly in a peak period. This is a candidate cause of over-coverage on E and was not tested separately.

A. Reproducibility and checks

Item	Value
Pre-registered configuration	config/date_model_e1.yaml v3, SHA-256 ae6a4a46…20152
Freeze record	FREEZE.json, SHA-256 3b62af1a…328dd; 9 runs. The final E1 scoring run came after it (E had been examined in Stage 7)
Code at scoring time	combined fingerprint 6eac633d…af40; a full copy is kept in snapshots/src_at_e1_scoring

Item	Value
Automated tests	317 pass, including synthetic end-to-end tests showing that scoring stops before any E output if the freeze, a model file, the sample, the calibration shifts or (after the post-hoc hardening) any input content changed, or if the post-hoc addendum is malformed or bound to a different freeze. The defect tests were also run against the earlier code to show that they failed there
Independent recomputation	every reported number recomputed from the saved per-order files with plain NumPy / pandas: 43 / 43 for E results, 43 / 43 for supplementary results
Post-hoc hardening	content hashes of all inputs, added after scoring as a labelled addendum bound to the freeze record's hash; the original freeze record is unchanged
Reproduction guide	docs/REPRODUCE_E1.md, recipes A–C; erratum of 2026-10-21 corrects the path used by recipe B (the historical code must sit at src/)
Process record	docs/E1_DECISIONS.md logs every change to the plan with its time, its reason and whether it came from the author or from a review
Environment	Python 3.13.16; numpy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1, scipy 1.18.1
Data	public Olist e-commerce dataset (Kaggle); raw files and order-level marts are not redistributed
Not done	a full rebuild from the raw CSVs by an independent reviewer; a second evaluation period; a registered test of the integer-aware shift or of recalibration that tracks changing conditions

B. Contribution statement (draft — the author must check every line)

Part	Who	Author to confirm
Reframing the question from late-order prediction to promise setting	Author, through the project and discussions	[ ]
Experiment scope and constraints (three model families, no tuning on E, staged checkpoints, disclosure rule)	Author	[ ]
Pre-registration text, code for selection, freeze and scoring, tests, simulation, post-hoc checks	Drafted and implemented with an AI assistant (Claude), revised over several review rounds	[ ]
Reviews that found design and code issues (comparison confound, freeze gaps, rounding, evaluation window, wording)	External review rounds [author: name the source]	[ ]
Libraries	scikit-learn (QuantileRegressor, HistGradientBoostingRegressor, mean_pinball_loss definition), NumPy, pandas, SciPy	—
Results I recomputed or re-derived myself	[author: list, e.g. the integer-day derivation, the V check, one E comparison]	[ ]

The lines above should be filled from the actual records, not from memory: docs/E1_DECISIONS.md and the review reports and evidence packages kept with it. The author should be able to explain, without notes: the integer-day derivation; why the calibration shift uses the (n + 1)q-th residual; why rounding after the shift favours an integer rule; why C* and B3 are the same model; and why HGB, the sharper model, ended up with the longer calibrated promise.